ViChan Technology (OPC) Private Limited

Transforming Learning into Career Opportunities.

Web Development | Mobile App Development | Software Solutions

Software Training | Internships

INTERNSHIP REPORT

Internship Title : Data Analytics

Project Title : E-Commerce Sales Analysis

Internship Duration

25-05-2026 to 25-06-2026

Submitted By :

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Academic Year	:	**2023 – 2026**

Internship Code : VCT/INT/MAY2026

Internship Period : May 2026 – June 2026

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1. INTRODUCTION

Data Analytics is the process of examining, transforming, and interpreting data to discover useful information, draw conclusions, and support decision-making. In today's digital world, organizations generate massive amounts of data through business transactions, customer interactions, websites, and mobile applications. Data Analytics helps organizations convert this raw data into meaningful insights that improve business performance and customer satisfaction.

The purpose of this internship was to gain practical knowledge of Data Analytics concepts and tools. During the internship, various stages of the analytics process such as data collection, data cleaning, data transformation, exploratory data analysis, visualization, and reporting were studied and implemented using Python libraries such as NumPy, Pandas, and Matplotlib.

The internship also provided hands-on experience in analyzing e-commerce sales data. Sales analytics plays a vital role in understanding purchasing behavior, customer preferences, product demand, and revenue trends. The project helped in identifying patterns within customer transactions and generating valuable business insights.

The internship bridged the gap between theoretical knowledge and practical industry applications by providing exposure to real-world datasets and analytical techniques.

2. COMPANY PROFILE

About the Company

ViChan Technology is an online training and software development organization that provides industry-oriented training programs and internship opportunities for students. The company focuses on developing technical skills and practical knowledge in various domains including Web Development, Mobile App Development, Data Analytics, Artificial Intelligence, and Software Engineering.

Vision

To empower students and professionals with practical industry skills through quality training and real-world project experience.

Mission

- To provide industry-relevant training programs.
- To enhance employability through practical projects.
- To bridge the gap between academic learning and industry requirements.
- To encourage innovation and technology adoption among students.

Services Offered

- Software Development
- Web Development
- Mobile App Development
- Online Internship Programs
- Data Analytics Training
- Web Development Training
- Mobile App Development Training
- Artificial Intelligence Training
- Project Guidance and Mentoring

Internship Program

The internship program focused on Data Analytics concepts and practical implementation using Python and data visualization tools. Participants were trained in handling datasets, performing analysis, and creating meaningful visual reports.

3. OBJECTIVES OF INTERNSHIP

The primary objectives of this internship were:

- To understand the fundamentals of Data Analytics.
- To learn the complete data analysis workflow.
- To gain practical experience in data cleaning and preprocessing.
- To analyze real-world e-commerce datasets.
- To learn NumPy for numerical computations.
- To use Pandas for data manipulation and analysis.
- To create visualizations using Matplotlib.
- To generate meaningful reports from data.
- To understand sales and customer behavior through analytics.
- To develop problem-solving and analytical skills.

4. TECHNOLOGIES LEARNED

Python Programming

Python is a popular programming language widely used in Data Analytics because of its simplicity and extensive library support.

Features:

- Easy syntax
- Large community support
- Rich data analytics ecosystem

NumPy

NumPy is a numerical computing library used for:

- Array creation
- Mathematical operations
- Statistical calculations
- Data manipulation

Pandas

Pandas is used for:

- Reading datasets
- Data cleaning
- Filtering data
- Sorting data

Important Functions:

- read_csv()
- head()
- tail()
- info()
- describe()
- groupby()
- merge()

Matplotlib

Matplotlib is used for data visualization.

Charts Created:

- Line Charts
- Bar Charts
- Pie Charts
- Histograms
- Scatter Plots

Jupyter Notebook

Used for:

- Writing Python code
- Data analysis
- Visualization
- Documentation

5. DAILY ACTIVITIES

Week 1

Day 1

- Internship Orientation
- Introduction to Data Analytics

Day 2

- Python Basics
- Variables and Data Types

Day 3

- Lists, Tuples, Dictionaries
- Control Statements

Day 4

- Functions and Modules
- Python Practice Programs

Day 5

- Introduction to NumPy
- Array Operations

Week 2

Day 6

- NumPy Statistical Functions

Day 7

- Introduction to Pandas

Day 8

- Reading CSV Files

Day 9

- Data Cleaning Techniques

Day 10

- Data Filtering and Sorting

Week 3

Day 11
- GroupBy Operations

Day 12
- Data Aggregation

Day 13
- Matplotlib Introduction

Day 14
- Bar Charts and Pie Charts

Day 15
- Histograms and Scatter Plots

Week 4

Day 16
- Project Dataset Collection

Day 17
- Data Cleaning

Day 18
- Sales Analysis

Day 19
- Visualization Development

Day 20
- Final Report Preparation

6. PROJECT WORK

Project Title

E-Commerce Sales Analysis

Problem Statement

E-commerce businesses generate large volumes of customer transaction data. Analyzing this data helps identify sales patterns, popular product categories, revenue trends, and customer purchasing behavior across regions and time periods.

Objective

- Analyze monthly sales revenue trends from 2020 to 2024.
- Identify top-performing product categories and brands.
- Examine order status breakdown and fulfillment rates.
- Determine payment method preferences among customers.
- Identify top cities and countries by revenue.
- Generate visual reports using Matplotlib.

Dataset Description

The dataset contains:

- Customer ID
- Customer Name
- Product Category
- Brand
- Product Name
- Quantity
- Unit Price
- Purchase Date
- Order Status
- Payment Method
- City
- Country

Software Requirements

Hardware

- Computer/Laptop
- Minimum 4GB RAM

Software

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib

Methodology

Step 1: Data Collection

- Dataset loaded from amazon.csv file.

Step 2: Data Cleaning

- Missing value removal
- Duplicate record removal
- Data type conversion

Step 3: Data Processing

- Total Sales calculation
- Average Sales
- Revenue by category and brand

Step 4: Data Analysis

- Product Performance
- Monthly Revenue Trends
- Country and City-wise Sales

Step 5: Visualization

- Monthly Sales Line Chart
- Order Status Pie Chart
- Payment Method Pie Chart
- Top Brands Bar Chart
- Top Categories Bar Chart
- Top Cities Bar Chart
- Country Revenue Bar Chart

Source Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
pd.set_option('display.max_colwidth', None)

amazon = pd.read_csv("amazon.csv")
print(amazon.shape)
print(amazon.head())
print(amazon.info())
print(amazon.isnull().sum())
print(amazon.duplicated().sum())

amazon.dropna(inplace=True)
amazon.drop_duplicates(inplace=True)

amazon['Purchase Date'] = pd.to_datetime(amazon['Purchase Date'])
amazon['Month'] = amazon['Purchase Date'].dt.to_period('M')
amazon['Total Price'] = amazon['Quantity'] * amazon['Unit Price']

# Monthly Sales Revenue
monthly_sales = amazon.groupby('Month')['Total Price'].sum().reset_index()
monthly_sales['Month'] = monthly_sales['Month'].astype(str)
plt.figure(figsize=(18, 4))
plt.plot(monthly_sales['Month'], monthly_sales['Total Price'], color='blue')
plt.title('Monthly Sales Revenue (2020-2024)')
plt.xlabel('Month')
plt.ylabel('Total Revenue')
plt.xticks(rotation=90)
plt.tight_layout()
plt.savefig('monthly_sales.png')
plt.show()

# Order Status Breakdown
order_status = amazon['Order Status'].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(order_status, labels=order_status.index, autopct='%1.1f%%', startangle=140)
plt.title('Order Status Breakdown')
plt.savefig('order_status.png')
plt.show()

# Payment Method Breakdown
payment = amazon['Payment Method'].value_counts()
plt.figure(figsize=(6, 6))
plt.pie(payment, labels=payment.index, autopct='%1.1f%%', startangle=140)
plt.title('Payment Method Breakdown')
plt.savefig('payment_methods.png')
plt.show()

# Top 5 Brands by Revenue
top_brands = amazon.groupby('Brand')['Total Price'].sum().nlargest(5).reset_index()
plt.figure(figsize=(8, 5))
plt.bar(top_brands['Brand'], top_brands['Total Price'], color='green')
plt.title('Top 5 Brands by Revenue')
```

```
plt.xlabel('Brand')
plt.ylabel('Total Revenue')
plt.savefig('top_brands.png')
plt.show()

# Top 5 Categories by Revenue
top_cat = amazon.groupby('Category')['Total Price'].sum().nlargest(5).reset_index()
plt.figure(figsize=(8, 5))
plt.bar(top_cat['Category'], top_cat['Total Price'], color='orange')
plt.title('Top 5 Categories by Revenue')
plt.xlabel('Category')
plt.ylabel('Total Revenue')
plt.savefig('top_categories.png')
plt.show()

# Top 5 Cities by Revenue
top_cities = amazon.groupby('City')['Total Price'].sum().nlargest(5).reset_index()
plt.figure(figsize=(8, 5))
plt.bar(top_cities['City'], top_cities['Total Price'], color='red')
plt.title('Top 5 Cities by Revenue')
plt.xlabel('City')
plt.ylabel('Total Revenue')
plt.savefig('top_cities.png')
plt.show()

# Revenue by Country
top_countries = amazon.groupby('Country')['Total Price'].sum().nlargest(5).reset_index()
plt.figure(figsize=(8, 5))
plt.bar(top_countries['Country'], top_countries['Total Price'], color='purple')
plt.title('Revenue by Country')
plt.xlabel('Country')
plt.ylabel('Total Revenue')
plt.savefig('top_countries.png')
plt.show()
```

Results

Findings include:

- Monthly sales revenue fluctuated between ~1.35M and 1.65M across 2020-2024.
- 74.6% of orders were successfully delivered; only 3% cancelled or returned.
- Credit Card is the most preferred payment method at 35%.
- Electronics is the top revenue-generating category.
- United States contributes the highest revenue among all countries.
- Charlotte, Dallas, San Jose, Seattle, and Philadelphia are the top revenue cities.
- CoreTech, KiddoFun, ReadMore, UrbanStyle, and Zenith are the top 5 brands.

Screenshots / Output Charts

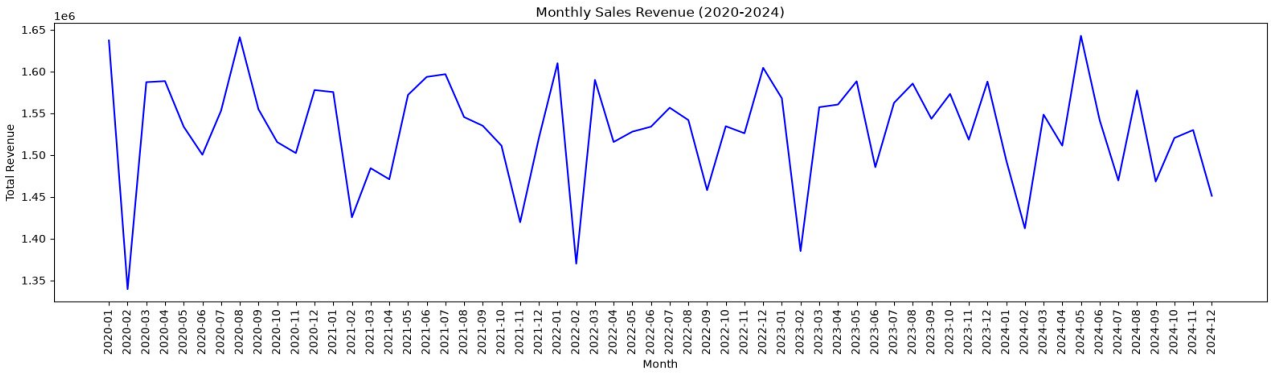


Figure 1: Monthly Sales Revenue (2020–2024)

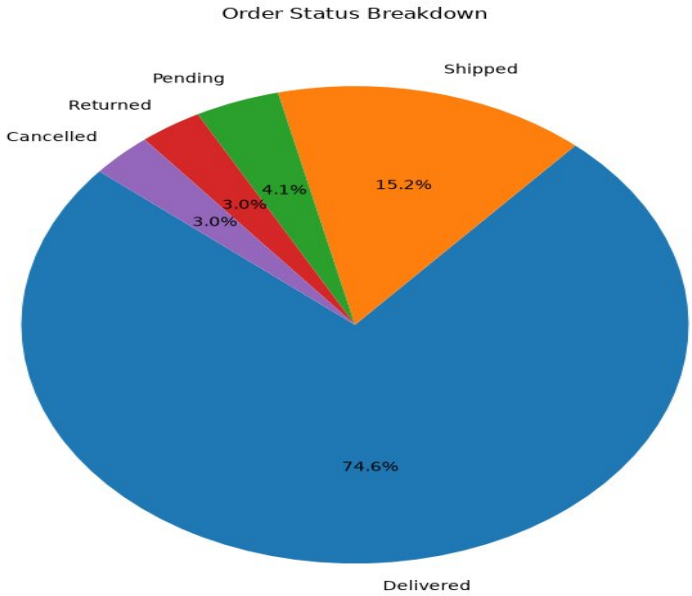


Figure 2: Order Status Breakdown

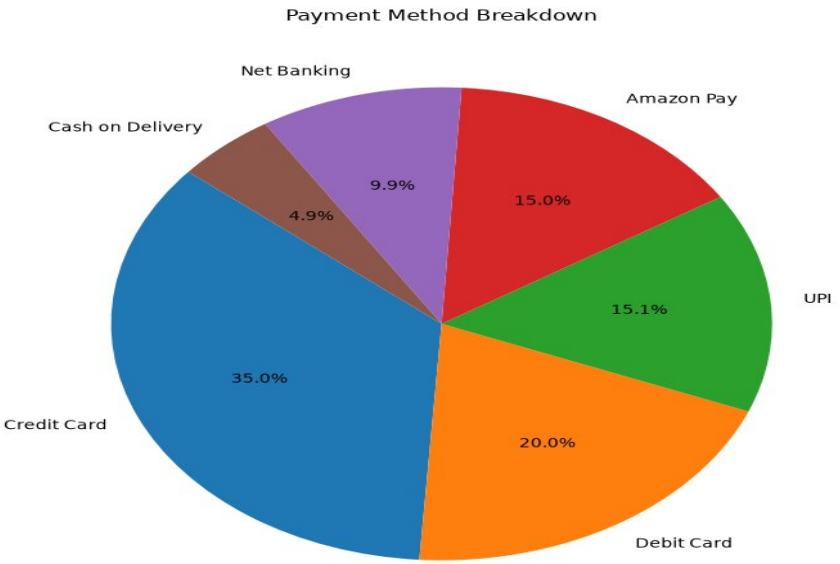


Figure 3: Payment Method Breakdown

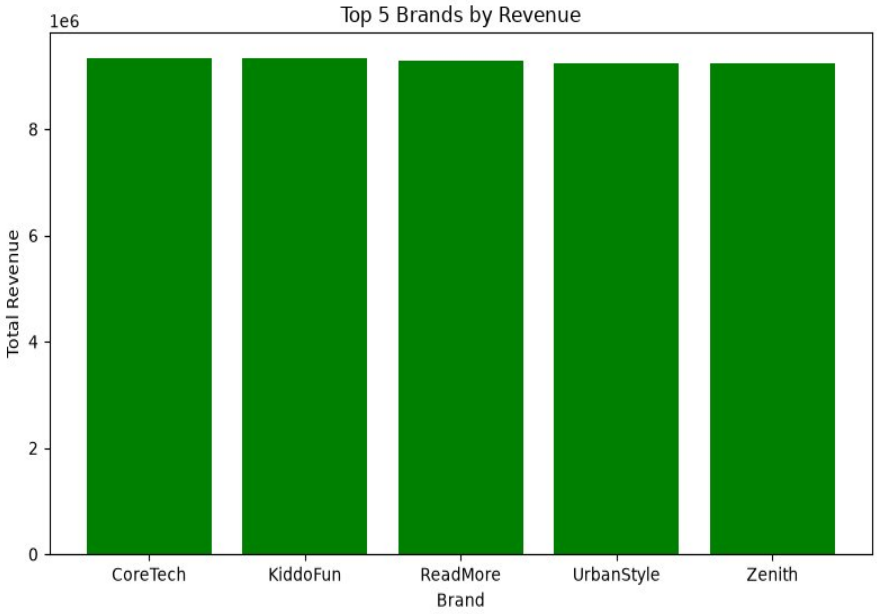


Figure 4: Top 5 Brands by Revenue

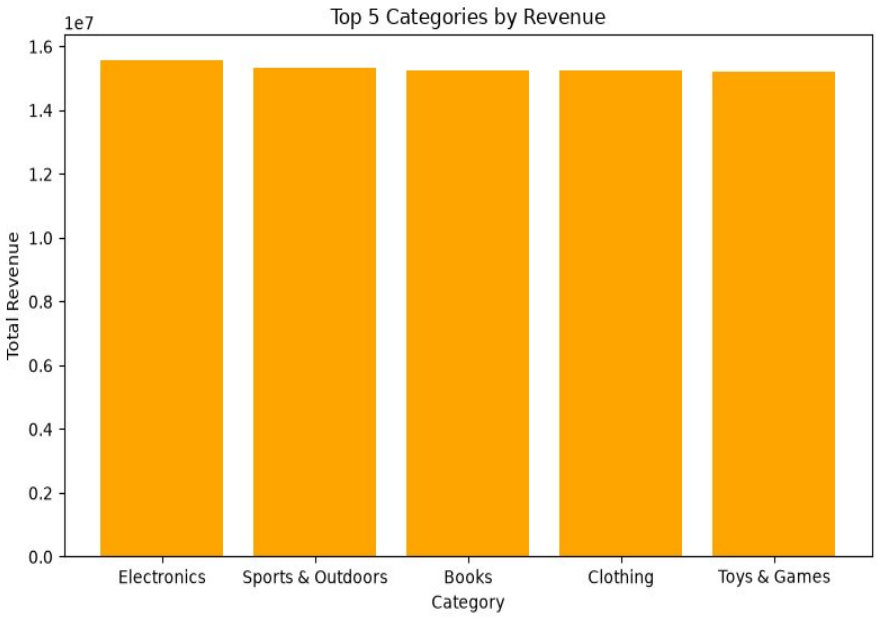


Figure 5: Top 5 Categories by Revenue

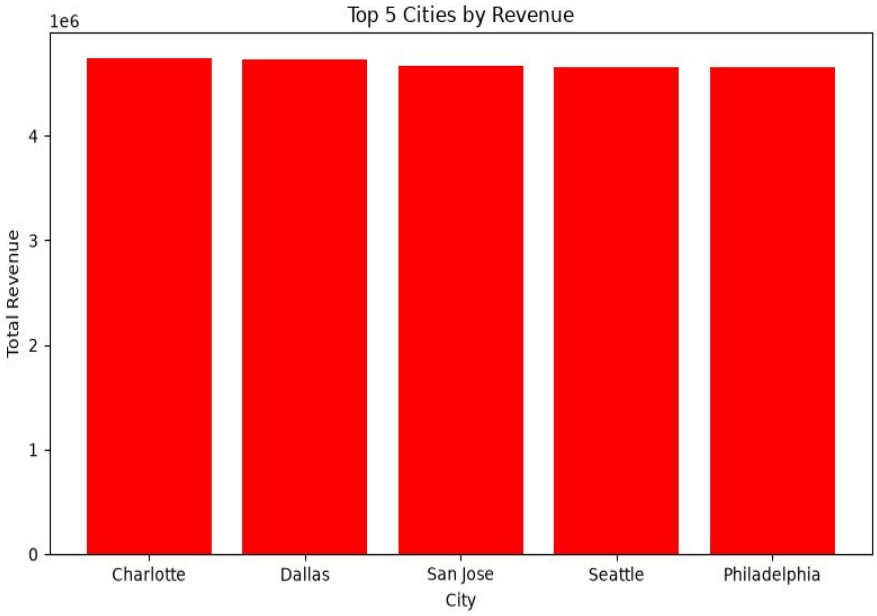


Figure 6: Top 5 Cities by Revenue

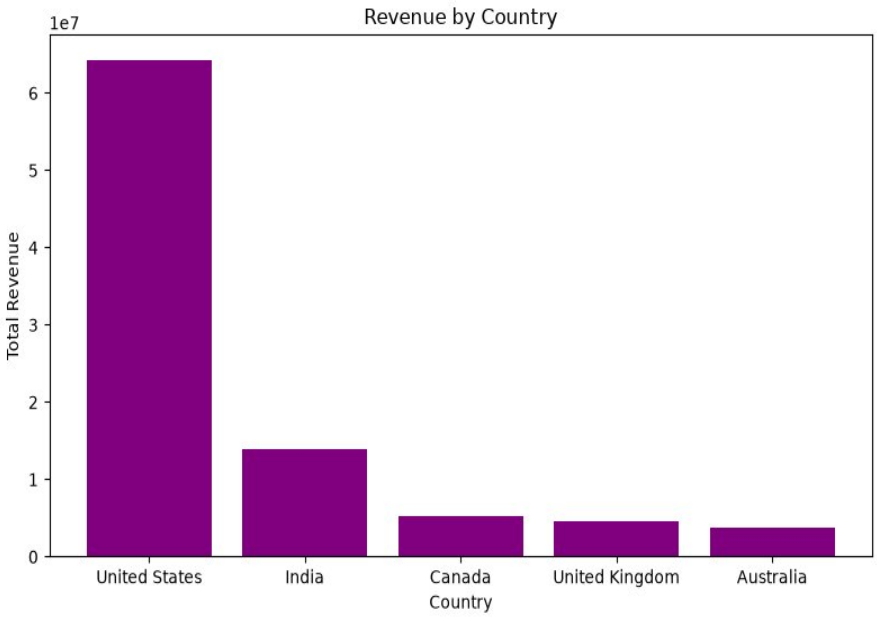


Figure 7: Revenue by Country

7. SKILLS GAINED

Technical Skills

- Python Programming
- NumPy Operations
- Pandas Data Analysis
- Data Cleaning
- Data Visualization
- Report Generation

Soft Skills

- Communication Skills
- Time Management
- Problem Solving
- Analytical Thinking
- Team Collaboration

8. CHALLENGES FACED

During the internship, several challenges were encountered:

Missing Data
Some records contained missing values.

Solution: Applied data cleaning techniques using Pandas dropna().

Duplicate Records
Duplicate entries affected accuracy.

Solution: Removed duplicates using drop_duplicates() in Pandas.

Data Visualization
Selecting appropriate charts for analysis was challenging.

Solution: Used different chart types and compared outputs.

Large Dataset Handling
Processing large data required optimization.

Solution: Used efficient Pandas groupby and aggregation operations.

9. OUTCOMES AND LEARNING EXPERIENCE

The internship provided practical exposure to Data Analytics concepts and tools.

Major learning outcomes:

- Understanding the analytics lifecycle.
- Working with real-world e-commerce datasets.
- Performing data cleaning and transformation.
- Creating visual dashboards and charts.
- Interpreting business insights from sales data.
- Improving analytical thinking and reporting skills.

The project enhanced technical skills and improved confidence in handling data-driven tasks.

10. CONCLUSION

The Data Analytics internship was highly beneficial in understanding the practical implementation of analytics techniques. Through the E-Commerce Sales Analysis project, valuable experience was gained in data collection, cleaning, processing, analysis, and visualization.

The internship strengthened programming skills, analytical capabilities, and business understanding. The knowledge acquired during this internship will be useful for future academic projects and professional career opportunities in Data Analytics and Data Science.

11. REFERENCES

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3. Pandas Documentation
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https://matplotlib.org

5. Kaggle Datasets
https://www.kaggle.com

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https://www.w3schools.com

7. Data Analytics Course Materials Provided During Internship

8. Jupyter Notebook Documentation
https://jupyter.org